

# 5

## Conventional Resists: Exposure and Bake Chemistry

Photoresists work by converting a spatial distribution of energy (the projected image of a photomask) into a spatial distribution of solubility of the resist in a developer. The first step in this chemical process is exposure, where a light-sensitive component of the resist forms a latent image in response to the aerial image. For so-called ‘conventional’ resists used for g-line and i-line exposures, the exposure products directly change the solubility of the resist. For a chemically amplified resist (discussed in the next chapter), a post-exposure bake is used to thermally induce a second chemical reaction, the product of which changes the solubility of the resist. In both cases, understanding the nature of the image transfer from a light distribution to a chemical distribution requires an understanding of the chemical kinetics of each of the reactions.

While this chapter deals with conventional resists, which are used only for the older g-line and i-line lithography processes, virtually all of the topics discussed here apply to chemically amplified resists as well. In a sense, chemically amplified resists are a technological superset of conventional resists. Thus, the discussion of chemically amplified resists in the next chapter will assume a familiarity with the material in this chapter.

### 5.1 Exposure

The kinetics of photoresist exposure is intimately tied to the phenomenon of absorption. The discussion below begins with a description of absorption, followed by the photochemical kinetics of exposure.

#### 5.1.1 Absorption

The first law of photochemistry (also called the Grotthuss–Draper law) states quite simply that only light that is absorbed is effective at causing chemical change. Thus a

photochemical reaction, such as the exposure of photoresist, must necessarily begin with the absorption of light. The phenomenon of absorption can be viewed on a macroscopic or a microscopic scale. On the macroscopic level, absorption is described by the familiar Lambert and Beer laws, which give a linear relationship between absorbance and path length multiplied by the concentration of the absorbing species. On the microscopic level, a photon is absorbed by an atom or molecule, promoting an electron to a higher energy state. Both methods of analysis yield useful information needed to describe the effects of light on a photoresist.

The basic law of absorption is an empirical one called Lambert's law. It was expressed by Johann Heinrich Lambert in 1760, though it was in fact first proposed some years earlier by Pierre Bouguer and is sometimes called the Lambert–Bouguer law. It can be expressed in differential form as

$$\frac{dI}{dz} = -\alpha I \quad (5.1)$$

where  $I$  is the intensity of light traveling in the  $z$ -direction through a medium, and  $\alpha$  is the absorption coefficient of the medium and has units of inverse-length. This law is basically a single photon absorption probability equation: the probability that a photon will be absorbed over some small distance traveled is proportional to the photon flux (i.e. the intensity). In a homogeneous medium (i.e.  $\alpha$  is not a function of  $z$ ), Equation (5.1) may be integrated to yield

$$I(z) = I_0 e^{-\alpha z} \quad (5.2)$$

where  $z$  is the distance the light has traveled through the medium and  $I_0$  is the intensity at  $z = 0$ . Note that the integrated form of the Lambert's law requires a homogeneous material. If the medium is inhomogeneous, Equation (5.2) becomes

$$I(z) = I_0 e^{-Abs(z)} \quad (5.3)$$

where  $Abs(z) = \int_0^z \alpha(z') dz' = \text{the absorbance}$

When working with electromagnetic radiation, it is often convenient to describe the radiation by its complex electric field vector. The propagation of an electric field through some material can implicitly account for absorption by using a complex index of refraction  $\mathbf{n}$  for the material such that

$$\mathbf{n} = n + i\kappa \quad (5.4)$$

The imaginary part of the index of refraction  $\kappa$  is related to the absorption coefficient by

$$\alpha = 4\pi\kappa/\lambda \quad (5.5)$$

where  $\lambda$  is the vacuum wavelength of the propagating light. Note that the sign of the imaginary part of the index in Equation (5.4) depends on the sign convention chosen for the phasor representation of the electric field (see Chapter 2).

In 1852, August Beer showed that for dilute solutions of an absorbing material in a nonabsorbing solvent, the absorption coefficient is proportional to the concentration of the absorbing species in the solution:

$$\alpha_{\text{solution}} = a_i C_i \quad (5.6)$$

where  $a_i$  is the molar absorption coefficient (sometimes called the extinction coefficient) of some absorbing species labeled  $i$  (given by  $a_i = \alpha_i MW/\rho$ , where  $\alpha_i$  is the absorption coefficient of the pure material,  $MW$  is its molecular weight,  $\rho$  is its density) and  $C_i$  is the concentration. The stipulation that the solution be dilute expresses a fundamental limitation of Beer's law. At high concentrations, where absorbing molecules are close together, the absorption of a photon by one molecule may be affected by a nearby molecule. Since this interaction is concentration dependent, it causes deviation from the linear relation of Equation (5.6). Also, an apparent deviation from Beer's law occurs if the real part of the index of refraction changes appreciably with concentration. Thus, the validity of Beer's law should always be verified experimentally over the concentration range of interest.

While generally described for solutions, Beer's law can also be applied to solid mixtures. For an  $N$  component homogeneous solid, the overall absorption coefficient becomes

$$\alpha_T = \sum_{i=1}^N a_i C_i \quad (5.7)$$

The linear addition of absorption terms presumes that Beer's law holds across components, i.e. that the absorption by one material is not influenced by the presence of the other materials. Of the total amount of light absorbed, the fraction of light that is absorbed by component  $i$  is given by

$$\frac{I_{Ai}}{I_{AT}} = \frac{a_i C_i}{\alpha_T} \quad (5.8)$$

where  $I_{AT}$  is the total light absorbed by the film, and  $I_{Ai}$  is the light absorbed by component  $i$ .

We will now apply the concepts of macroscopic absorption to a typical positive photoresist. For g-line (436nm) and i-line (365nm) lithography, the most common resists are of the diazonaphthoquinone/novolac variety. These positive photoresists are made up of three major components: a base novolac resin that gives the resist its structural properties and etch resistance; a photoactive compound (PAC; this is the light-sensitive component in the resist, also known as the *sensitizer*) called a diazonaphthoquinone (DNQ); and a solvent that renders these components into liquid form for spin coating. (Although photoresist drying during post-apply bake is intended to drive off solvents, a resist may contain up to 10% solvent after a typical post-apply bake – see section 5.2.2.) Denoting the resin  $R$ , the sensitizer  $M$  and the solvent  $S$ , a fourth component appears during exposure: exposure products  $P$  generated by the reaction of  $M$  with ultraviolet (UV) light.



Applying Beer's law, the absorption coefficient  $\alpha$  is then

$$\alpha = a_M M + a_P P + a_R R + a_S S \quad (5.10)$$

Since  $P$  is being produced and  $M$  is disappearing through the exposure process, this equation makes it clear the  $\alpha$  will be changing during exposure. If  $M_0$  is the initial sensitizer concentration (i.e. with no exposure), the stoichiometry of the exposure reaction gives

$$P = M_0 - M \quad (5.11)$$

Equation (5.10) may be rewritten as

$$\alpha = Am + B \quad (5.12)$$

where  $A = (a_M - a_P) M_0$

$$B = a_P M_0 + a_R R + a_S S$$

$m = M/M_0$  = the relative sensitizer concentration.

$A$  and  $B$  are called the bleachable and nonbleachable absorption coefficients, respectively, and make up the first two Dill photoresist parameters.<sup>1</sup> Other nonbleachable components of the photoresist (such as a dye additive, leveling agents to reduce striations during spin coating, etc.) are added to the  $B$  term above. Note that when  $A > 0$ , the photoresist will become more transparent as it is exposed (that is, the photoresist ‘bleaches’). If  $A$  is negative, the resist will darken upon exposure.

The quantities  $A$  and  $B$  are experimentally measurable and can be easily related to typical resist absorbance curves, measured using a UV spectrophotometer. When the resist is fully exposed,  $M = 0$  and

$$\alpha_{\text{exposed}} = B \quad (5.13)$$

Similarly, when the resist is unexposed,  $m = 1$  ( $M = M_0$ ) and

$$\alpha_{\text{unexposed}} = A + B \quad (5.14)$$

From this,  $A$  may be found by

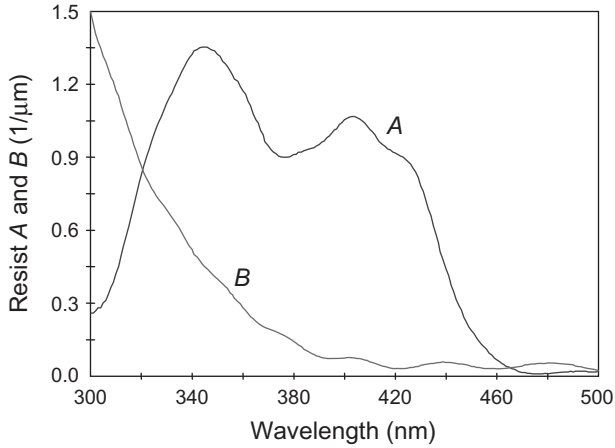
$$A = \alpha_{\text{unexposed}} - \alpha_{\text{exposed}} \quad (5.15)$$

Thus,  $A(\lambda)$  and  $B(\lambda)$  may be determined from the UV absorbance curves of unexposed and completely exposed resist (Figure 5.1). A more complete description of the measurement of  $A$  and  $B$  will be given at the end of this chapter.

As mentioned previously, Beer’s law is empirical in nature and consequently should be verified experimentally. In the case of positive photoresists, this means formulating resist mixtures with differing sensitizer to resin ratios and measuring the resulting  $A$  parameters. Previous work has shown that Beer’s law was valid for a typical conventional photoresist over the full practical range of sensitizer concentrations.<sup>2</sup>

## 5.1.2 Exposure Kinetics

The kinetics of exposure is best understood by considering absorption on the microscopic scale. On a microscopic level, the absorption process can be thought of as photons being absorbed by an atom or molecule causing an outer electron to be promoted to a higher energy state. The absorption of a photon can have three possible results: the absorbed energy can be dissipated as heat (lattice vibrations); it can be re-emitted as another photon (fluorescence); or it can lead to a chemical reaction. It is the third outcome



**Figure 5.1** Resist parameters *A* and *B* as a function of wavelength measured with a UV spectrophotometer for a typical g-line resist (a 5-arylsulfonate DNQ)

that is important for the sensitizer since it is the absorption of UV light that leads to the chemical conversion of *M* to *P* as seen in Equation (5.9). Einstein first quantified absorption on a microscopic (quantum) scale in 1916–1917 by using probabilities of possible absorption events. Here, we will ignore stimulated absorption by assuming a reasonably low light intensity so that at any given time relatively few molecules will be in the excited state.

Consider a substance *M* being exposed to quasi-monochromatic radiation. Since the speed of light is constant in a given material, the Lambert law can be converted to a kinetic absorption equation by letting  $dz = vdt$  where  $v$  is the speed of light (the rate at which new photons arrive). The rate of photon absorption by the molecule *M* is proportional to  $N_M$  (the number density of *M*), the number of photons, and the molecule's absorption cross section  $\sigma_{M-abs}$ :

$$-\frac{d\phi_M}{dt} = vN_M\sigma_{M-abs}\phi_M \quad (5.16)$$

where  $\phi_M$  is the number density of photons. The intensity *I* is an energy flux – the amount of energy per unit area per second – and can be related to the photon flux (the photon density times the speed the photons are traveling) multiplied by the energy of one photon:

$$I = v\phi_M \left( \frac{hc}{\lambda} \right) \quad (5.17)$$

where *h* is Planck's constant and *c* is the speed of light in vacuum. Einstein's relation for photon absorption is

$$-\frac{d\phi_M}{dt} = N_M\sigma_{M-abs} \left( \frac{\lambda}{hc} \right) I = N_MB_E I/v \quad (5.18)$$

where  $B_E$  is Einstein's coefficient of absorption for material  $M$  (the probability per unit time per unit energy density of the radiation field that the photon will be absorbed by a molecule of  $M$ , sometimes written as  $B_{12}$ ). Note that  $I/\nu$  represents the energy density of the photons.

In order to relate Equation (5.18) to measurable quantities, one must relate the microscopic theory to macroscopic observations. The absorption cross section of the molecule  $M$  can be related to its molar absorptivity by

$$\sigma_{M-\text{abs}} = \frac{\alpha_M}{N_M} = \frac{a_M}{N_A} \quad (5.19)$$

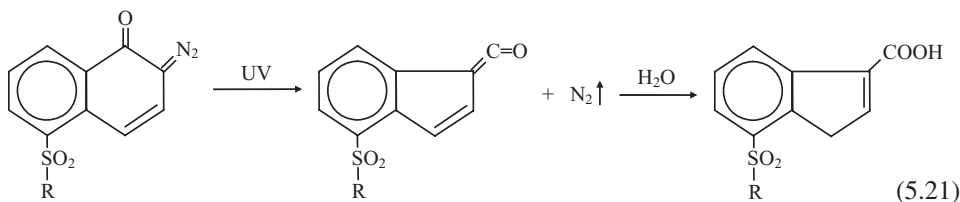
where  $N_A$  is Avogadro's number ( $6.022 \times 10^{23}$  atoms/mol). Einstein's coefficient of absorption for material  $M$  is the speed of light times the absorption cross section divided by the energy of one photon:

$$B_E = \nu \sigma_{M-\text{abs}} \left( \frac{\lambda}{hc} \right) = \frac{\sigma_{M-\text{abs}}}{n} \left( \frac{\lambda}{h} \right) = \frac{a_M \lambda}{n N_A h} \quad (5.20)$$

where  $n$  is the refractive index of the resist.

As an example of the use of this microscopic absorption analysis, consider a typical g-line resist of density 1.2 g/ml. For a PAC with molecular weight 400 g/mol formulated at 30% by weight, the initial PAC concentration will be 0.9 molar (moles/liter). Since  $a_p$  is very small for these resists,  $a_M$  is about equal to the Dill bleachable absorption coefficient divided by the initial PAC concentration. For  $A = 0.63 \mu\text{m}^{-1}$ , this gives  $a_M = 7 \times 10^6 \text{ cm}^2/\text{mol}$ . The resulting PAC absorption cross section is about  $0.12 \text{ \AA}^2$ . Since the energy of one photon at this wavelength is  $4.56 \times 10^{-19} \text{ J}$ , the resulting Einstein coefficient of absorption is about  $4.7 \times 10^5 \text{ m}^3/\text{J}\cdot\text{s}$ .

The actual chemistry of diazonaphthoquinone exposure is given by<sup>3</sup>



where  $R$  is a moderate to large molecular weight ballast group (in some cases  $R$  is connected to the polymer resin). The DNQ converts to an indene intermediate before converting to its final product, a carboxylic acid. It is interesting to note that nitrogen gas is given off (it diffuses quickly out of the film during exposure) and that water is required for this reaction. Water is supplied by the humidity in the atmosphere, which is one reason why clean room humidity levels are tightly controlled (usually at a set point of around 40% relative humidity). If the relative humidity of the clean room environment drops below about 30%, the above DNQ exposure reaction will not go to completion. Instead, the intermediate ketene will slowly react with the novolac resin to form an ester (more on this in section 5.2.1).

The chemical reaction in Equation (5.9) can be rewritten in a more general form as



where  $M$  is the sensitizer,  $M^*$  is the sensitizer molecule in an excited state,  $P$  is the indene carboxylic acid (product), and  $k_1$ ,  $k_2$ ,  $k_3$  are the rate constants for each reaction. Simple kinetics can now be applied. The proposed mechanism in Equation (5.22) assumes that all reactions are first order. Thus, the rate equation for each species can be written.

$$\begin{aligned}\frac{dM}{dt} &= k_2 M^* - k_1 M \\ \frac{dM^*}{dt} &= k_1 M - (k_2 + k_3) M^* \\ \frac{dP}{dt} &= k_3 M^*\end{aligned}\quad (5.23)$$

A system of three coupled linear first-order differential equations can be solved exactly using Laplace transforms and the initial conditions

$$\begin{aligned}M(t=0) &= M_0 \\ M^*(t=0) &= P(t=0) = 0\end{aligned}\quad (5.24)$$

However, if one uses the steady-state approximation, the solution becomes much simpler. This approximation assumes that in a very short time the excited molecule  $M^*$  comes to a steady state, i.e.  $M^*$  is formed as quickly as it disappears. In mathematical form,

$$\frac{dM^*}{dt} = 0 \quad (5.25)$$

The intermediate  $M^*$  does indeed come to a steady state quickly, on the order of  $10^{-8}$  seconds or faster.<sup>4</sup> Thus,

$$\frac{dM}{dt} = -KM \quad (5.26)$$

where  $K = \frac{k_1 k_3}{k_2 + k_3}$

The overall rate constant  $K$  is proportional to the intensity of the exposure radiation. The rate constant  $k_1$  will be proportional to the rate of photon absorption (by the Grotthuss–Draper law), which in turn is proportional to the photon flux (by Lambert's law), and thus the intensity. A more useful form of Equation (5.26) is then

$$\frac{dm}{dt} = -CIm \quad (5.27)$$

where the relative sensitizer concentration  $m (= M/M_0)$  has been used and  $C$  is the standard exposure rate constant and the third Dill photoresist parameter (collectively, the three Dill parameters are also called the *ABC parameters*). A more thorough microscopic analysis of the exposure process allows this exposure rate constant to be broken down into the product of the absorption cross section of the sensitizer and the quantum yield of the reaction (the fraction of absorbed photons that produce the chemical change,  $\Phi$ ):

$$C = \Phi \sigma_{M-\text{abs}} \left( \frac{\lambda}{hc} \right) = \frac{\Phi B_E}{\nu} = \frac{\Phi a_M \lambda}{N_A hc} \quad (5.28)$$

Maximum resist sensitivity comes from high quantum efficiency, all absorption coming from the sensitizer, and an absorption coefficient adjusted to be one over the thickness of the resist (see Problem 5.2).

A solution to the exposure rate Equation (5.27) is simple if the intensity within the resist is constant throughout the exposure (i.e. when  $A = 0$ ). Integrating Equation (5.27) gives

$$m = e^{-CIt} \quad (5.29)$$

where  $Dose = \int_0^t I dt = It$  for the case of constant intensity during the exposure process.

This result illustrates an important property of first-order kinetics called *reciprocity*. The amount of chemical change is controlled by the product of light intensity and exposure time. Doubling the intensity and cutting the exposure time in half will result in the exact same amount of chemical change. This product of intensity and exposure time is called the *exposure dose* or *exposure energy*. This idea can be made more usefully explicit by relating the exposure dose at some point inside the resist to the exposure dose incident on the resist. For either the simple case of absorption as shown in Equation (5.2) or (5.3), or for the more general case of standing waves in the resist film as described in Chapter 4, the intensity in the resist  $I(z)$  is always directly proportional to the incident intensity  $I_{inc}$ . Thus,

$$I(x, y, z) = I_{inc} I_r(x, y, z) \quad (5.30)$$

where  $I_r$  is the relative intensity, that is, the intensity in the resist assuming a unit incident intensity. Equation (5.29) can now be rewritten as

$$m(x, y, z) = \exp[-CDose_{inc} I_r(x, y, z)] \quad (5.31)$$

where  $Dose_{inc}$  is the incident exposure dose. Reciprocity exists whenever all instances of exposure time and incident intensity can be replaced by incident dose in an expression for the final latent image  $m(x, y, z)$ .

Reciprocity failures can occur when mechanisms other than the first-order kinetics shown here become significant. For example, an insufficient supply of water during exposure can lead to reciprocity failure if high intensity exposures use up water in the resist film faster than water from the atmosphere can diffuse back into the photoresist. Reciprocity failure can also result from high intensities that heat up the resist during exposure, causing a temperature-dependent change in the molar absorptivity or the quantum yield.

For many resists, especially the conventional DNQ resists popular for g-line and i-line exposure, the intensity within the resist is not constant during exposure. In fact, many resists *bleach* upon exposure, that is, they become more transparent as the sensitizer  $M$  is converted to product  $P$ . This corresponds to a positive value of  $A$ , as seen, for example, in Figure 5.1. Since the intensity varies as a function of exposure time, this variation must be known in order to solve the exposure rate equation. In the simplest possible case, a resist film coated on a substrate of the same index of refraction, only absorption affects the intensity within the resist. Thus, Lambert's law of absorption, coupled with Beer's law, could be applied:

$$\frac{dI}{dz} = -(Am + B)I \quad (5.32)$$

where Equation (5.12) was used to relate the absorption coefficient to the relative sensitizer concentration. Equations (5.29) and (5.32) are coupled, and thus become first-order nonlinear partial differential equations that must be solved simultaneously. Equations (5.29) and (5.32) were first solved analytically by Herrick:<sup>5</sup>

$$z = \int_{m(z=0)}^{m(z)} \frac{dy}{y[A(1-y) - B \ln(y)]} \quad (5.33)$$

where  $y$  is a dummy variable for the purposes of integration, and

$$I(z) = I(0) \frac{A[1 - m(z)] - B \ln[m(z)]}{A[1 - m(0)] - B \ln[m(0)]} \quad (5.34)$$

The sensitizer concentration at the top of the resist,  $m(0)$ , is obtained from the nonbleaching solution [Equation (5.29)] since the intensity at the top of a bleaching resist on a nonreflecting substrate does not change during exposure. For many g-line and i-line resists,  $A \gg B$ . Thus, at least near the beginning of exposure,  $B$  can often be safely neglected (i.e.  $Am \gg B$ ). Under these conditions, the integral in Equation (5.33) can be solved exactly (by setting  $B = 0$ ) giving

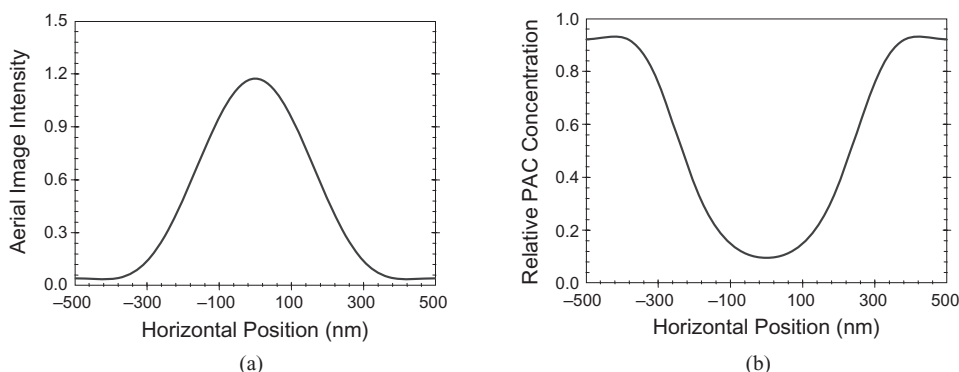
$$\begin{aligned} m(z) &= \frac{m(0)}{m(0) + [1 - m(0)]e^{-Az}} \\ \frac{I(z)}{I(0)} &= \frac{1 - m(z)}{1 - m(0)} = \frac{e^{-Az}}{m(0) + [1 - m(0)]e^{-Az}} = \frac{m(z)}{m(0)} e^{-Az} \end{aligned} \quad (5.35)$$

Although an analytical solution exists for the simple problem of exposure with absorption only, in more realistic problems the variation of intensity with depth in the film is more complicated than Equation (5.32). In fact, the general exposure situation results in the formation of standing waves, as discussed in Chapter 4. Thus, numerical solutions to the time-varying intensity and chemical composition of the film during exposure will almost certainly be required whenever a bleaching photoresist film is used.

The final result of exposure is the conversion of an image intensity in resist  $I_r(x, y, z)$  into a latent image  $m(x, y, z)$ . Figure 5.2 illustrates a one-dimensional case.

## 5.2 Post-Applied Bake

Baking a resist may have many purposes, from removing solvent to causing chemical amplification. In addition to the intended results, baking may also cause numerous unintended outcomes. For example, the light-sensitive component of the resist may decompose at temperatures typically used to remove solvent. The solvent content of the resist can impact diffusion and amplification rates for a chemically amplified resist. Also, all aspects of baking will probably affect the dissolution properties of the resist. Baking a photoresist remains one of the most complicated and least understood steps in the lithographic process.

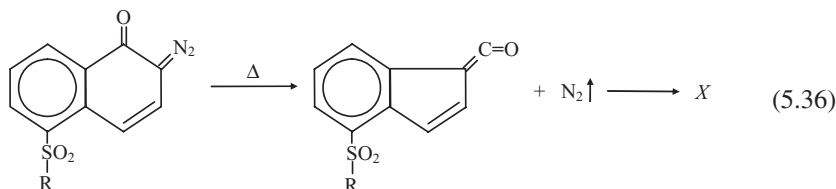


**Figure 5.2** The exposure process takes an aerial image (a) and converts it into a latent image (b)

The post-apply bake (PAB) process, also called a softbake or a prebake, involves drying the photoresist after spin coating by removing excess solvent. There are four major effects of removing solvent from a photoresist film: (1) film thickness is reduced and stabilized (i.e. becomes relatively independent of the time spent waiting for exposure and development); (2) post-exposure bake and development properties are changed; (3) adhesion is improved; and (4) the film becomes less tacky and thus less susceptible to particulate contamination. Unfortunately, there are other consequences of baking many photoresists. For conventional resists, the photosensitive component may begin to decompose at temperatures greater than about  $70^{\circ}\text{C}$ <sup>6</sup> (for chemically amplified resists, decomposition begins at much higher temperatures). Thus, one must search for the optimum post-apply bake conditions that will maximize the benefits of solvent evaporation while maintaining a tolerable level of sensitizer decomposition.

### 5.2.1 Sensitizer Decomposition

When heated to temperatures above about  $70\text{--}80^{\circ}\text{C}$ , the sensitizer of a DNQ-type positive photoresist begins to decompose to a nonphotosensitive product. The initial reaction mechanism is thought to be identical to that of the sensitizer reaction during ultraviolet exposure.<sup>6,7</sup>



The possible identity of the product or products  $X$  will be discussed below.

To determine the concentration of sensitizer  $M$  as a function of post-apply bake time and temperature, consider the first-order decomposition reaction,



If we let  $M'_0$  be the concentration of sensitizer before post-apply bake and  $M_0$  the concentration of sensitizer after post-apply bake, simple kinetics tells us that

$$\begin{aligned} \frac{dM_0}{dt} &= -K_T M_0 \\ M_0 &= M'_0 \exp(-K_T t_b) \\ m' &= e^{-K_T t_b} \end{aligned} \quad (5.38)$$

where  $t_b$  is the post-apply bake time,  $K_T$  is the decomposition rate constant at the absolute temperature  $T$  and  $m' = M_0/M'_0$ , the fraction of sensitizer remaining after the bake. The dependence of  $K_T$  upon temperature can be described by the Arrhenius equation,

$$K_T = A_r e^{-E_a/RT} \quad (5.39)$$

where  $A_r$  is the Arrhenius coefficient,  $E_a$  is the activation energy of the reaction,  $R$  is the universal gas constant (1.98717 cal/mole-K or 8.31431 J/mole-K), and  $T$  is the absolute temperature (K). Thus, the two parameters  $E_a$  and  $A_r$  allow us to know  $m'$  as a function of the post-apply bake conditions, provided Arrhenius behavior is followed. In polymer systems, caution must be exercised since bake temperatures near the glass transition temperature sometimes lead to non-Arrhenius behavior. For normal post-apply bakes of typical photoresists, the Arrhenius model appears well founded.

The effect of this decomposition is a change in the chemical makeup of the photoresist. Thus, any parameters which are dependent upon the quantitative composition of the resist are also dependent upon post-apply bake. The most important of these parameters fall into three categories: (1) optical (exposure) parameters such as the resist absorption coefficient; (2) diffusion parameters during post-exposure bake; and (3) development parameters such as the development rates of unexposed and completely exposed resist. A technique will be described to measure  $E_a$  and  $A_r$  and thus begin to quantify these effects of post-apply bake.

In the Dill model of exposure described in the previous section, the exposure of a positive photoresist can be characterized by the three parameters  $A$ ,  $B$  and  $C$ , as given in Equations (5.12) and (5.29). These expressions do not explicitly take into account the effects of post-apply bake on the resist composition. To do so, we can modify Equation (5.12) to include absorption by the component  $X$ :

$$B = a_p M_0 + a_R R + a_X X \quad (5.40)$$

where  $a_X$  is the molar absorption coefficient of the decomposition product  $X$  (and the absorption term for the solvent has been neglected for simplicity). The stoichiometry of the decomposition reaction gives

$$X = M'_0 - M_0 \quad (5.41)$$

Thus,

$$B = a_p M_0 + a_R R + a_X (M'_0 - M_0) \quad (5.42)$$

Let us consider two cases of interest, no bake (NB) and full bake (FB). When there is no post-apply bake (meaning no decomposition),  $M_0 = M'_0$  and

$$\begin{aligned} A_{\text{NB}} &= (a_M - a_P)M'_0 \\ B_{\text{NB}} &= a_P M'_0 + a_R R \end{aligned} \quad (5.43)$$

We shall define 'full bake' as a post-apply bake which decomposes all of the sensitizer. Thus  $M_0 = 0$  and

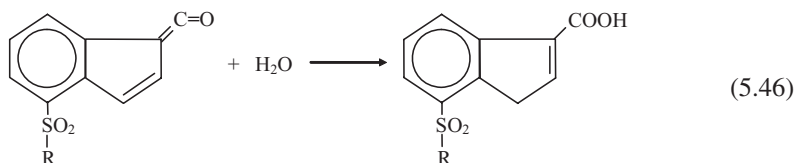
$$\begin{aligned} A_{\text{FB}} &= 0 \\ B_{\text{FB}} &= a_X M'_0 + a_R R \end{aligned} \quad (5.44)$$

Using these special cases in our general expressions for  $A$  and  $B$ , we can show explicitly how these two parameters vary with sensitizer decomposition:

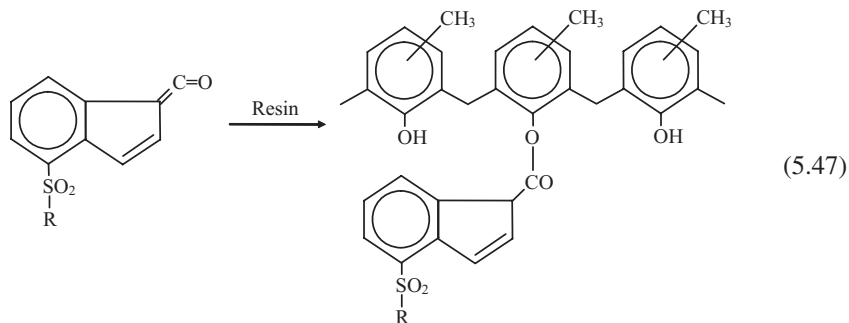
$$\begin{aligned} A &= A_{\text{NB}} m' \\ B &= B_{\text{FB}} - (B_{\text{FB}} - B_{\text{NB}}) m' \end{aligned} \quad (5.45)$$

The  $A$  parameter decreases linearly as decomposition occurs, and  $B$  typically increases slightly (since  $B_{\text{FB}}$  is typically slightly higher than  $B_{\text{NB}}$ ).

The development rate, as we shall see in Chapter 7, is dependent on the concentration of PAC in the photoresist. However, the product  $X$  can also have a large effect on the development rate. There are two likely products and the most common outcome of a post-apply bake decomposition is a mixture of the two. The first product is formed via the reaction below and is identical to the product of UV exposure.



As can be seen, this reaction requires the presence of water. A second reaction, which does not require water, is the esterification of the ketene with the novolac resin.



Both possible products have a dramatic effect on dissolution rate. The carboxylic acid is very soluble in developer and enhances dissolution. The formation of carboxylic acid can be thought of as a blanket exposure of the resist. The dissolution rate of unexposed resist ( $r_{\text{min}}$ ) will increase due to the presence of the carboxylic acid. The dissolution rate of fully exposed resist ( $r_{\text{max}}$ ), however, will not be affected. Since the chemistry of the dissolution process is unchanged, the basic shape of the development rate function will also remain

unchanged. The ester, on the other hand, is very difficult to dissolve in aqueous solutions and thus retards the dissolution process. It will have the effect of decreasing  $r_{\max}$ , although the effects of ester formation on the full dissolution behavior of a resist are not well known.

If the two mechanisms given in Equations (5.46) and (5.47) are taken into account, the rate Equation (5.38) will become

$$\frac{dM_0}{dt} = -k_1 M_0 - k_2 [\text{H}_2\text{O}] M_0 \quad (5.48)$$

where  $k_1$  and  $k_2$  are the rate constants of Equations (5.46) and (5.47), respectively. For a given concentration of water in the resist film,  $[\text{H}_2\text{O}]$ , this reverts to Equation (5.38) where

$$K_T = k_1 + k_2 [\text{H}_2\text{O}] \quad (5.49)$$

Thus, the relative importance of the two reactions will depend not only on the ratio of the rate constants but on the amount of water in the resist film. The concentration of water is a function of atmospheric conditions during the bake and the past history of the resist-coated wafer.

Examining Equation (5.45), one can see that the parameter  $A$  can be used as a means of measuring  $m'$ , the fraction of sensitizer remaining after post-apply bake. Thus, by measuring  $A$  as a function of post-apply bake time and temperature, one can determine the activation energy and the corresponding Arrhenius coefficient for the proposed decomposition reaction. Figure 5.3a shows the variation of the resist parameter  $A$  with post-apply bake conditions for one particular resist.<sup>8</sup> According to Equations (5.38) and (5.45), this variation should take the form

$$\begin{aligned} \frac{A}{A_{\text{NB}}} &= e^{-K_T t_b} \\ \ln \left( \frac{A}{A_{\text{NB}}} \right) &= -K_T t_b \end{aligned} \quad (5.50)$$

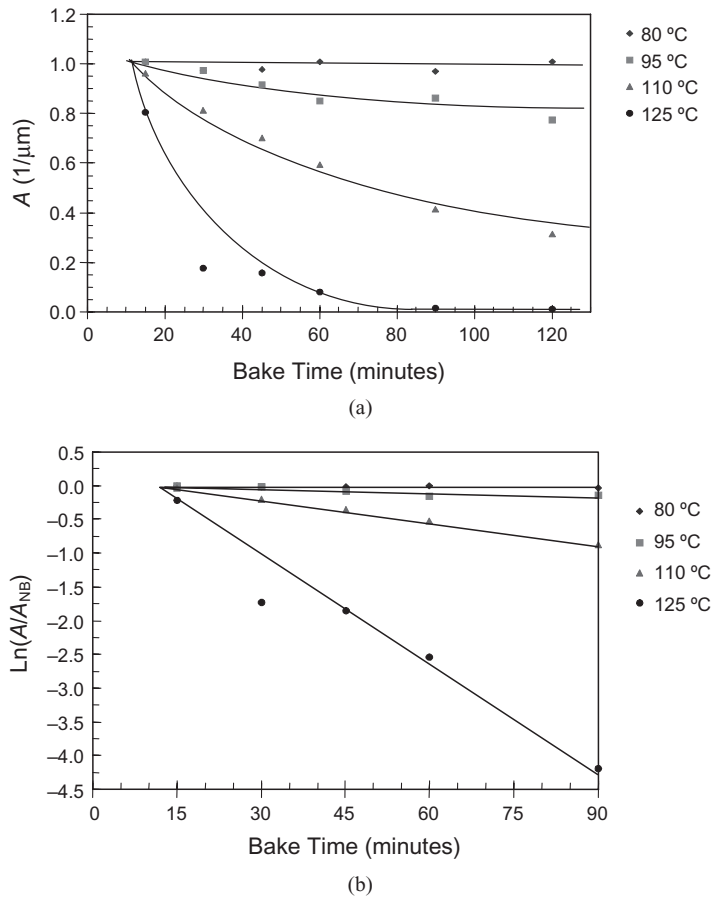
Thus, a plot of  $\ln(A)$  versus bake time should give a straight line with a slope equal to  $-K_T$ . This plot is shown in Figure 5.3b. Knowing  $K_T$  as a function of temperature, one can determine the activation energy and Arrhenius coefficient from Equation (5.39). One should note that the parameters  $A_{\text{NB}}$ ,  $B_{\text{NB}}$  and  $B_{\text{FB}}$  are wavelength dependent, but  $E_a$  and  $A_i$  are not.

Figure 5.3 shows an anomaly in which there is a lag time before decomposition occurs. This lag time is the time it took the wafer and wafer carrier to reach the temperature of the convection oven. Equation (5.38) can be modified to accommodate this phenomena,

$$m' = e^{-K_T(t_b - t_{\text{wup}})} \quad (5.51)$$

where  $t_{\text{wup}}$  is the warm-up time. A lag time of about 11 minutes was observed when convection oven baking a 1/4"-thick glass substrate in a wafer carrier. For a silicon wafer on a proximity hot plate, the warm-up time is on the order of 15–20 seconds.

From the data presented in Figure 5.3b, the activation energy is 30.3 Kcal/mol and the natural logarithm of the Arrhenius coefficient (in 1/minute) is 35.3. Thus, a 100°C,



**Figure 5.3** The variation of the resist absorption parameter  $A$  with post-apply bake time and temperature for Kodak 820 resist at 365 nm (a convection oven bake was used): (a) linear plot, and (b) logarithmic plot

30-minute convection oven post-apply bake would decompose 11% of the photoactive compound. The nature of the decomposition product (the identity of  $X$ ) will also affect the value of  $B_{FB}$ . If thermal decomposition leads to generation of both carboxylic acid and ester, only the ester will affect the change of  $B$  from  $B_{NB}$  to  $B_{FB}$ . In this case, Equation (5.44) must be modified to take into account the fraction of PAC that decomposes into ester,  $\phi$ , by replacing  $a_X$  with  $a_X\phi + (1 - \phi)a_P$ .

The examples given above for photoactive compound decomposition all involve DNQ/novolac resist systems used at g-line and i-line exposure wavelengths. For these systems, typical bake temperatures always result in some (hopefully small) amount of sensitizer decomposition. However, for chemically amplified resists, the photoacid generators are much more thermally stable. Typical post-apply bake temperatures result in essentially no thermal decomposition of the chemically amplified sensitizer.

### 5.2.2 Solvent Diffusion and Evaporation

Thermal processing of photoresists (post-apply bake and post-exposure bake) can dramatically influence resist performance in a number of ways. It is well known that residual solvent has a powerful influence on the dissolution rate of thin polymer films and that post-apply bake (PAB) determines the resist's residual solvent content. However, quantitative determination of the influence of bake conditions on residual solvent is difficult. In order to describe solvent evaporation during the baking of a photoresist, a model for how the diffusivity varies during the bake (that is, as a function of the changing photoresist film composition) must be established. In this section, a historical review of the Fujita–Doolittle equation for solvent diffusivity is given. Then, modifications to this equation are proposed to improve its accuracy and dynamic range as needed for its use in describing PAB.

The influence of temperature, molecular weight and chemical composition on the physical and mechanical properties of polymers is a difficult but important topic of study. Since the early 1950s the use of a 'free-volume' description of physical relaxation phenomena in polymers and other materials has proven extremely useful. Although the idea that the free, unoccupied volume in a liquid or solid greatly influences the physical properties of the material is quite old, it was first applied successfully by Arthur Doolittle in 1951.<sup>9</sup>

Doolittle described the influence of temperature on the viscosity of a liquid in two steps: the viscosity depends on the free volume in the liquid, and this free volume depends on temperature. The fraction of the volume of a liquid which is free, that is, unoccupied by the molecules of the liquid, was expressed as

$$v_f = \frac{V - V_0}{V} \quad (5.52)$$

where  $v_f$  is the free volume fraction,  $V$  is the volume that a specified mass of the liquid occupies at some temperature  $T$ , and  $V_0$  is the volume occupied by the liquid extrapolated to zero absolute temperature assuming no phase change. The assumption here is that the 'liquid' extrapolated to absolute zero will contain no free volume and that the thermal expansion of the liquid is due completely to the creation of free volume. Although the first assumption seems quite reasonable, the second is probably less accurate. Thermal expansion of the liquid will undoubtedly lead to free volume formation, but the increased vibrations of the liquid molecules will also lead to an increased 'occupied' volume, the volume that is occupied by the molecule to the exclusion of other molecules. Doolittle's free volume expression can be modified by multiplying by  $\zeta$ , the fraction of the increased volume which is actually free. Thus,

$$v_f = \zeta \frac{V - V_0}{V} \quad (5.53)$$

where the magnitude of  $\zeta$  for a liquid is probably close to unity, but for a solid could be considerably smaller.

The difficulty in measuring the free volume is in determining the value of  $V_0$ . Doolittle attempted this by measuring the density of a liquid as a function of temperature, fitting this data to an empirical equation, then extrapolating to zero temperature. With this

empirically determined temperature dependence for the free volume of the liquid, Doolittle then showed that experimental viscosity data as a function of temperature behaved as

$$\eta = Ae^{B/v_f} \quad (5.54)$$

where  $\eta$  is the viscosity and  $A$  and  $B$  are empirical constants. Equation (5.54) was found to match the experimental data over a wider range of temperatures much better than any of the other empirical expressions in common use at the time, lending credence to this free volume interpretation. Interestingly, the experimental value of  $B$  obtained by Doolittle was very close to unity (0.9995) and the free volume of the liquid ranged from 0.22 to 0.68 over the temperatures studied.

Others began applying the ideas of free volume to polymer systems. Describing both viscosity and diffusion processes in polymers by the same mechanism, Bueche<sup>10</sup> calculated the probability that sufficient free volume would be available to allow movement of the polymer. Relaxation processes (i.e. processes that are limited by the movement of the polymer) were thought to include such phenomena as viscous flow and diffusion. Thus, describing the rate at which a polymer moved was the first step in defining other relaxation rates. Since the polymer segments are vibrating as a function of temperature, the volume occupied by the polymer will have a thermodynamically controlled probability distribution about the average volume per polymer segment. If the volume occupied by the polymer segment exceeds some critical volume, the polymer segment can move or 'jump' to a new configuration. Thus, by integrating the volume probability from this critical volume to infinity, the frequency of jumps can be determined. In turn, the viscosity is almost completely controlled by this jump frequency. By comparing this theory with experimental polymer viscosity data, the temperature dependence of the average volume of a polymer segment was deduced.

For temperatures above the glass transition temperature of the polymer, Bueche described the polymer volume by

$$V = V_{T_g}(1 + (\alpha_1 + \alpha_2)T) \quad (5.55)$$

where  $V_{T_g}$  is the volume at the glass transition temperature,  $\alpha_1$  is the coefficient of thermal expansion of the solid-like polymer below the glass transition temperature, and  $\alpha_1 + \alpha_2$  is the thermal expansion coefficient of the liquid-like polymer above the glass transition temperature. This abrupt change in the coefficient of thermal expansion is considered one of the fundamental indicators of the glass transition phenomenon. Both  $\alpha_1$  and  $\alpha_2$  typically have magnitudes on the order of  $5 \times 10^{-4} \text{ K}^{-1}$  for most polymers.

Fox and Flory measured the specific volume of polystyrene<sup>11</sup> and polyisobutylene<sup>12</sup> as a function of temperature for many molecular weights and found that Equation (5.55) was quite adequate. Bueche and Fox and Flory went on to speculate that of the total volume expansion given by Equation (5.55), only the  $\alpha_2$  term resulted in the generation of free volume. Essentially, they argued that  $\zeta$  of Equation (5.53) was very small for the solid-like behavior of the polymer, and that the excess volume that comes from thermal expansion above the glass transition temperature is all free volume. Thus,

$$\begin{aligned} v_f &= v_g, & T &\leq T_g \\ v_f &= v_g + \alpha_2(T - T_g), & T &\geq T_g \end{aligned} \quad (5.56)$$

where  $T_g$  is the glass transition temperature and  $v_g$  is the fractional free volume at  $T_g$ .

The power of the free volume approach to relaxation mechanisms is that the fundamental relationship of the relaxation mechanism to free volume is independent of the mechanism by which free volume changes. Fujita, Kishimoto and Matsumoto<sup>13</sup> used this fact to expand the Doolittle model to include the effect of small amounts of solvent in the polymer. Solvent dissolved in a polymer results in additional free volume which in turn increases diffusivity and decreases viscosity. Fujita *et al.* modified the free volume Equation (5.56) to include solvent content:

$$v_f = v_g + \alpha_2(T - T_g) + \beta\phi_s \quad (5.57)$$

where  $\phi_s$  is the fractional volume of solvent and  $\beta$  describes the fraction of this solvent volume which can be considered free. Here,  $T_g$  is usually considered the glass transition temperature of the completely dry (no dissolved solvent) polymer. For the polymethyl acrylate polymer and four different solvents studied by Fujita,  $\beta$  was found to be 0.19.

The Fujita–Doolittle equation has been used extensively to characterize the change in solvent diffusivity with changing solvent concentration. There are, however, constraints to this approach that have led some workers in this field to criticize its use and to look to alternate expressions. In particular, higher solvent concentration causes a dilution of the polymer-induced free volume with the addition of solvent. A correction for this effect gives<sup>14</sup>

$$v_f = (1 - \phi_s)v_g + \alpha_2(T - T_g) + \phi_s\beta \quad (5.58)$$

Equation (5.58) is a more accurate form of the Fujita free volume expression which is valid for higher concentrations of solvent, while still requiring the same number of parameters as the original free volume expression.

The final expression for free volume, Equation (5.58), can now be combined with the Doolittle equation to give a modified Fujita–Doolittle equation for the diffusivity  $D$  of a solvent in a polymer matrix as a function of temperature and solvent content.

$$D = Ae^{-B/v_f} \quad (5.59)$$

Letting  $D_0$  be the minimum diffusivity, that is the diffusivity in the limit of no solvent content and  $T = T_g$ ,

$$D_0 = Ae^{-B/v_g} \quad (5.60)$$

Thus, Equation (5.59) can be rearranged as

$$D = D_0 e^{-B(1/v_f - 1/v_g)}$$

where

$$v_f = (1 - \phi_s)v_g + \alpha_2(T - T_g) + \phi_s\beta \quad (5.61)$$

It will often be more convenient to describe the solvent content as a mass fraction rather than a volume fraction. Letting  $x$  be the mass fraction and  $\rho$  the density,

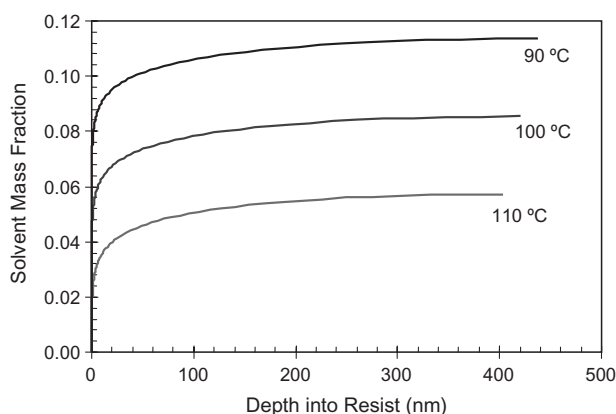
$$\begin{aligned}
 x_s &\approx \frac{\rho_s}{\rho_p} \phi_s \\
 v_f &= (1 - x_s)v_g + \alpha_2(T - T_g) + x_s\beta' \\
 \beta' &= \frac{\rho_p}{\rho_s}\beta + v_g\left(1 - \frac{\rho_p}{\rho_s}\right) \approx \frac{\rho_p}{\rho_s}\beta
 \end{aligned}
 \tag{5.62}$$

where the subscripts 's' and 'p' refer to solvent and polymer, respectively.

Based on previous work, the order of magnitude for each of the terms in Equation (5.62) is known. Doolittle and Williams *et al.*<sup>15</sup> found  $B$  to be approximately 1.0, while Fujita *et al.* found  $B = 0.73$ . Cohen and Turnbull<sup>16</sup> limit  $B$  to be between 0.5 and 1 (their so-called geometric factor). The Williams, Landel and Ferry (WLF) equation predicts  $v_g = 0.025$  and  $\alpha_2 = 4.8 \times 10^{-4} \text{ K}^{-1}$ , but further work by Ferry<sup>17</sup> found values of  $v_g/B$  between 0.013 and 0.07, with most of the data between 0.02 and 0.035, and values of  $\alpha_2$  between 1 and  $11 \times 10^{-4} \text{ K}^{-1}$ , with most data between 3 and  $5 \times 10^{-4} \text{ K}^{-1}$ . Fox and Flory found  $\alpha_2 = 3 \times 10^{-4} \text{ K}^{-1}$ . Fujita found  $\beta$  to be 0.19, and in any case it must be  $\leq 1.0$ . The value of  $\beta'$  should be similar. From these typical numbers and the diffusivity model given above, one can see that the diffusivity of solvent changes by four to six orders of magnitude during the course of a typical post-apply bake.

Using this solvent diffusion model, one can predict the impact of PAB time and temperature on the final resist thickness and the resulting solvent distribution as a function of depth into the resist at the end of the bake. Such modeling provides for a very interesting result. Any combination of initial resist thickness and initial solvent content that produces a certain final resist thickness always results in essentially the same solvent distribution in the resist at the end of the bake. Figure 5.4 shows the distribution of solvent left in the resist at the end of a 60-second bake of various temperatures using the parameters from Table 5.1.

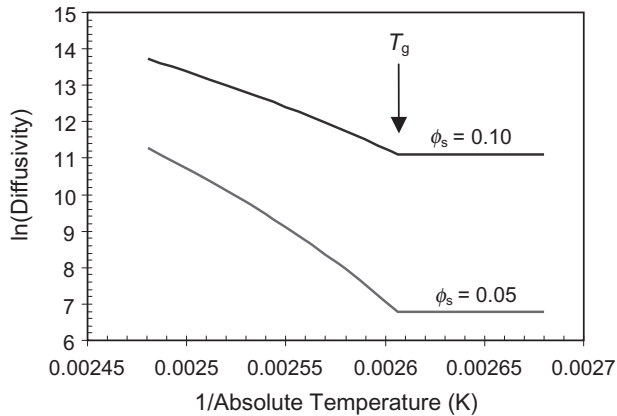
Another consequence of using the modified Fujita–Doolittle equation is a prediction of how solvent diffusivity varies with temperature. Using the parameters from Table 5.1,



**Figure 5.4** Predicted variation of solvent concentration as a function of depth into the resist at the end of a 60-second post-apply bake

**Table 5.1** Typical solvent diffusion parameters in photoresist

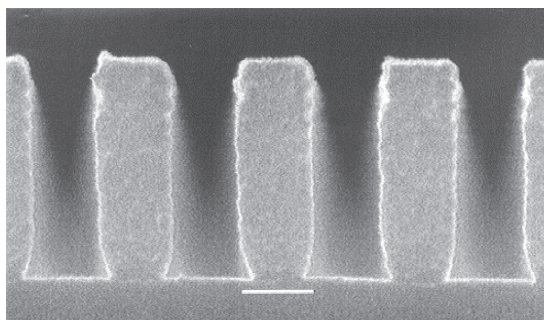
| Parameter                  | Value   |
|----------------------------|---------|
| $B$                        | 0.737   |
| $\beta'$                   | 0.351   |
| $v_g$                      | 0.0289  |
| $\alpha_2$ (1/K)           | 8.7E-04 |
| $T_g$ (°C)                 | 110.5   |
| $D_0$ (nm <sup>2</sup> /s) | 0.0967  |

**Figure 5.5** Temperature dependence of solvent diffusivity (using the parameters from Table 5.1 and assuming a solvent mass fractions of 0.05 and 0.1) showing an essentially fixed diffusivity below the glass transition temperature

and assuming a solvent mass fraction of 0.05, an Arrhenius plot of solvent diffusivity is given in Figure 5.5. There is a large discontinuity at the glass transition temperature, since the model assumes such a discontinuity as a consequence of Equation (5.56). Note that above the glass transition temperature, the behavior is non-Arrhenius, that is, the curve is not perfectly linear in the Arrhenius plot of Figure 5.5.

### 5.2.3 Solvent Effects in Lithography

Residual solvent in the film after baking can have two significant effects on the lithographic performance of a photoresist. Just as residual solvent increases the free volume and thus the diffusivity of solvent during PAB, any residual solvent at post-exposure bake will increase the diffusivity of exposure products. For a chemically amplified resist, especially when baking in a diffusion-controlled temperature regime, this increase in acid diffusivity can have a quite large impact. Figure 5.6 shows an example where the top portion of the resist exhibits standing waves while the rest of the resist does not. Reduced solvent content at the top of the resist leads to reduced acid diffusion during PEB and (as will be discussed in the following section) less smoothing out of the standing wave pattern.



**Figure 5.6** Lower solvent content at the top of this 248-nm resist leads to reduced acid diffusion during PEB, and thus the presence of standing waves only at the top of the resist (photo courtesy of John Petersen, used with permission)

**Table 5.2** Development  $r_{\max}$  and  $n$  values (see Chapter 7) fitted to measured dissolution rate data as a function of PAB temperature (and thus, solvent content) for a thick *i*-line resist<sup>18</sup>

| Temperature (°C) | wt.% Solvent | $r_{\max}$ (nm/s) | Develop $n$ |
|------------------|--------------|-------------------|-------------|
| 70               | 21.1         | 104.3             | 1.71        |
| 80               | 19.5         | 90.3              | 1.77        |
| 90               | 17.8         | 78.9              | 1.96        |

As Figure 5.5 showed, diffusivity grows almost exponentially as a function of the temperature above  $T_g$ . Thus, for the post-exposure bake to cause appreciable diffusion (as will be discussed in the next section), the PEB temperature must be above the  $T_g$  of the resist. However, as seen above, the  $T_g$  of a resist film is a function of the amount of solvent in the resist, and thus a function of the post-apply bake conditions. For a sufficiently long post-apply bake, the  $T_g$  of the resist film approaches the post-apply bake temperature. For shorter bakes, the resist film  $T_g$  will be somewhat below the PAB temperature. Thus, as a simple rule of thumb, the PEB temperature must be the same or greater than the PAB temperature in order for appreciable diffusion to occur during PEB.

Secondly, residual solvent can impact resist dissolution rates, affecting both the average dissolution rate and the variation of dissolution rate with exposure dose (Table 5.2). Unfortunately, this behavior is rarely well characterized for any given resist material.

### 5.3 Post-exposure Bake Diffusion

Many attempts have been made to reduce the standing wave effect and thus increase linewidth control and resolution. One particularly useful method is the post-exposure, predevelopment bake as described by Walker.<sup>19</sup> A 100°C oven bake for 10 minutes was found to reduce the standing wave ridges on a resist sidewall significantly. This phenomenon can be explained quite simply as due to the diffusion of sensitizer in the resist (which

acts as a dissolution inhibitor for a conventional resist, see Chapter 7) during the high-temperature bake. A mathematical model which predicts the results of such a post-exposure bake (PEB) is described below.

In general, molecular diffusion is governed by Fick's Second Law of Diffusion, which states (in one dimension,  $r$ )

$$\frac{\partial C_A}{\partial t} = \frac{\partial}{\partial r} \left( D_A \frac{\partial C_A}{\partial r} \right) \quad (5.63)$$

where  $C_A$  is the concentration of species A,  $D_A$  is the diffusion coefficient of A at some temperature  $T$ , and  $t$  is the time that the system is at temperature  $T$ . A more general form is

$$\frac{\partial C_A}{\partial t} = \nabla(D_A \nabla C_A) \quad (5.64)$$

In many cases, it is quite accurate to assume that the diffusivity is independent of concentration (and thus position). Under these conditions, this differential equation can be solved given a set of boundary conditions (two for each dimension) and an initial distribution of A.

Consider one possible initial condition known as the impulse source (a delta function of concentration). At some point  $x_0$ , there are  $N$  moles of substance A and at all other points there is no A. Thus, the concentration at  $x_0$  is infinite. For boundary conditions, assume zero concentration as  $r$  goes to  $\pm\infty$  for all time. Given this initial distribution of A and these boundary conditions, the solution to Equation (5.63) is the Gaussian distribution function,

$$C_A(x) = \frac{N}{\sqrt{2\pi\sigma^2}} e^{-r^2/2\sigma^2} \quad (5.65)$$

where  $\sigma = \sqrt{2D_A t}$  is the diffusion length, and  $r = x - x_0$ .

In practice there are no impulse sources. Instead, we can approximate an impulse source as having some concentration  $C_0$  over some small distance  $\Delta x$  centered at  $x_0$ , with zero concentration outside of this range. An approximate form of Equation (5.65) is then

$$C_A(x) \cong \frac{C_0 \Delta x}{\sqrt{2\pi\sigma^2}} e^{-r^2/2\sigma^2} \quad (5.66)$$

This solution is fairly accurate if  $\Delta x \ll \sigma$ . If there are two 'impulse' sources located at  $x_1$  and  $x_2$ , with initial concentrations  $C_1$  and  $C_2$  each over a range  $\Delta x$ , the concentration of A at  $x$  after diffusion is

$$C_A(x) = \left[ \frac{C_1}{\sqrt{2\pi\sigma^2}} e^{-r_1^2/2\sigma^2} + \frac{C_2}{\sqrt{2\pi\sigma^2}} e^{-r_2^2/2\sigma^2} \right] \Delta x \quad (5.67)$$

where  $r_1 = x - x_1$  and  $r_2 = x - x_2$ .

If there are a number of sources, Equation (5.67) becomes

$$C_A(x) = \frac{\Delta x}{\sqrt{2\pi\sigma^2}} \sum_n C_n e^{-r_n^2/2\sigma^2} \quad (5.68)$$

Extending the analysis to a continuous initial distribution  $C_0(x)$ , Equation (5.68) becomes

$$C_A(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \int_{-\infty}^{\infty} C_0(x') e^{-(x-x')^2/2\sigma^2} dx' \quad (5.69)$$

where  $x'$  is now the distance from the point  $x$ . Equation (5.69) is simply the convolution (denoted by  $\otimes$ ) of the original concentration profile with a Gaussian:

$$C_A(x) = C_0(x) \otimes f(x) \quad (5.70)$$

where  $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/2\sigma^2}$

This equation can now be made to accommodate two-dimensional diffusion:

$$C_A(x,y) = C_0(x,y) \otimes f(x,y) \quad (5.71)$$

where

$$f(x,y) = \frac{1}{2\pi\sigma^2} e^{-r^2/2\sigma^2}$$

$$r = \sqrt{x^2 + y^2}$$

Three-dimensional diffusion can similarly be calculated (as shown below).

From this formulation of diffusion, one can see that the diffusion impulse response function of Equation (5.65) or its higher dimensional equivalents is analogous to the imaging point spread function of Chapter 2. Thus,  $f(r)$  is sometimes referred to as the diffusion point spread function (*DPSF*).

We are now ready to apply our solution to the diffusion of sensitizer in a conventional photoresist during a post-exposure bake. For the full three-dimensional case, the sensitizer distribution after exposure can be described by  $m(x,y,z)$ , where  $m$  is the relative sensitizer concentration. According to Equation (5.71) the relative sensitizer concentration after a post-exposure bake,  $m^*(x,y,z)$ , is given by

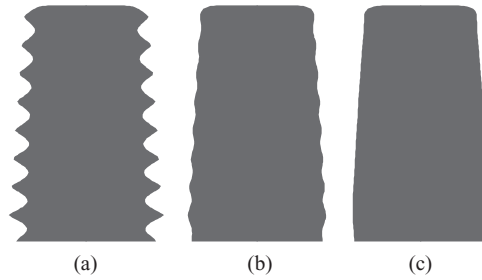
$$m^*(x,y,z) = \frac{1}{(2\pi\sigma^2)^{3/2}} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} m(x',y',z') e^{-((x-x')^2 + (y-y')^2 + (z-z')^2)/2\sigma^2} dx'dy'dz' \quad (5.72)$$

or,

$$m^*(x,y,z) = m(x,y,z) \otimes \text{DPSF} \quad (5.73)$$

For a given latent image, the only parameter that needs to be specified to solve Equation (5.72) is the diffusion length  $\sigma$ , or equivalently, the diffusion coefficient  $D_A$  and the bake time  $t$ . In turn,  $D_A$  is a function of the bake temperature  $T$  (by the Arrhenius equation for temperature ranges which do not traverse the glass transition temperature) and, of course, the resist system used. Unfortunately, it is quite difficult to measure the diffusivity of sensitizer in photoresist. From Walker's work, one can estimate the values of  $E_a$  and  $D_0$  for the resist studied to be about 35 Kcal/mol and  $3.2 \times 10^{21} \text{ nm}^2/\text{s}$ , respectively.

The diffusion model can now be used to simulate the effects of a post-exposure bake on standing waves for a conventional resist (Figure 5.7). Obviously, the major effect is



**Figure 5.7** Typical i-line photoresist profile simulations (using PROLITH) for resist on silicon as a function of the PEB diffusion length: (a) 20 nm, (b) 40 nm and (c) 60 nm

the smoothing out of the standing waves. Less obvious from these pictures is the gradual degradation of the photoresist image itself. The trade-off between these two effects puts a significant constraint on the acceptable values of the diffusion length. In order to remove standing waves, the diffusion length must be on the order of or larger than half of the standing wave period. But to avoid degrading the feature itself, diffusion length must be much smaller than the smallest feature size being printed.

From Chapter 4, recall that the standing wave period is  $\lambda/2n_{\text{resist}}$  where  $n_{\text{resist}}$  is the resist refractive index (the period will be slightly less than this for oblique illumination). For i-line exposure of a typical resist, this puts the half period at about 55 nm. Thus, diffusion lengths must be at about this magnitude or larger to completely remove standing waves. Since i-line resists are rarely used to resolve features less than about 300 nm, the requirement that diffusion length is much less than feature size is also realized. However, for 248-nm lithography this is no longer the case. The standing wave half period is about 35–40 nm, but features smaller than 130 nm are regularly imaged. Thus, to remove standing waves effectively, the diffusion length must approach one-third of the feature size, a fraction that is too large to avoid significant image degradation. As a result, for 248- and 193-nm lithography, standing waves are reduced using bottom antireflection coatings rather than relying solely on diffusion during PEB. Additionally, it is important to remember that a post-exposure bake, while capable of reducing or eliminating standing waves, will have no impact on the photoresist swing curve. Since bottom antireflection coatings can be very effective at reducing both standing waves and swing curves, they have become the preferred solution for critical lithography levels.

## 5.4 Detailed Bake Temperature Behavior

In the discussion of section 5.2.1, a somewhat clumsy approach of adding a ‘warm-up time’ to the bake analysis was used to account for the noninstantaneous heating of the resist. By modeling the heat transfer from the hot plate to the wafer, a more accurate and

detailed picture can be drawn of how the temperature of the wafer varies with time during the bake.

We can approximate the temperature profile of the resist on a typical wafer baking (track) system by the following differential equation (called a lumped capacitance model for heat transfer):<sup>20,21</sup>

$$\rho C_p L \frac{dT}{dt} = -\frac{k_{\text{air}}}{\delta} (T - T_{\text{plate}}) - h(T - T_{\text{ambient}}) \quad (5.74)$$

where  $\rho$  is the density of silicon,  $C_p$  is the specific heat capacity of silicon,  $L$  is the thickness of the wafer,  $T$  is the temperature of the resist-coated wafer,  $k_{\text{air}}$  is the thermal conductivity of air,  $\delta$  is the thickness of the gap between the hot plate and the wafer, and  $h$  is a heat transfer coefficient for the convective heat lost from the top surface of the wafer to the surroundings. In this model, the term on the left side represents the thermal mass of the wafer, and the terms on the right side represent heat transfer on the bottom side of the wafer from the hot plate and heat loss from the top of the wafer to the surroundings. Note that the thermal mass of the photoresist is extremely small compared to that of the wafer, so that it can be ignored. Also, the thermal conductivity of silicon is high enough that the temperature gradient from the bottom to the top of the wafer can also be ignored. Radiative heat transfer mechanisms are small enough to be ignored as well.

The solution to this lumped capacitance heat transfer model is

$$T = T^* - (T^* - T_{\text{initial}})e^{-t/\tau} \quad (5.75)$$

where  $T_{\text{initial}}$  is the initial temperature of the wafer,  $T^*$  is the equilibrium (infinite time) temperature given by

$$T^* = \frac{\frac{k_{\text{air}}}{\delta} T_{\text{plate}} + h T_{\text{ambient}}}{\frac{k_{\text{air}}}{\delta} + h} \quad (5.76)$$

and  $\tau$  is the time constant for heating (or possibly cooling) the wafer,

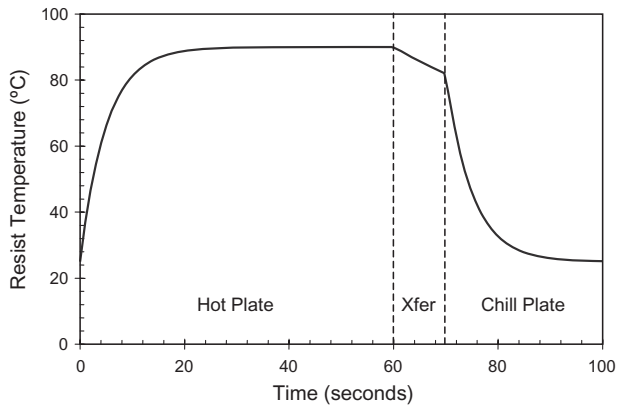
$$\tau = \frac{\rho C_p L}{\frac{k_{\text{air}}}{\delta} + h} \quad (5.77)$$

Some typical values for the constants in these thermal model equations are given in Table 5.3.

We can now describe the full temperature profile of the resist on a typical wafer track system with a three-stage model for heat transfer to the resist-coated wafer: a proximity hot plate bake as described above; a short time for transfer from the hot plate to a chill plate; and then proximity cooling on the chill plate. During the second stage where the wafer is transferred from the hot plate to the chill plate, the gap  $\delta$  is assumed to be very large so that the first term on the right side of Equation (5.74) is unimportant, and the second term on the right side is assumed to represent heat loss from the top

**Table 5.3** Typical values for silicon wafer properties

| Parameter                                                        | Value | Units               |
|------------------------------------------------------------------|-------|---------------------|
| $L$ (wafer thickness), 300-mm-diameter wafer                     | 0.775 | mm                  |
| $L$ (wafer thickness), 200-mm-diameter wafer                     | 0.725 | mm                  |
| $L$ (wafer thickness), 150-mm-diameter wafer                     | 0.675 | mm                  |
| $L$ (wafer thickness), 100-mm-diameter wafer                     | 0.525 | mm                  |
| $C_p$ (silicon molar heat capacitance) @ 25 °C                   | 19.8  | J/K-mole            |
| $C_p$ (silicon molar heat capacitance) @ 125 °C                  | 22.3  | J/K-mole            |
| $\rho$ (silicon density)                                         | 2.33  | g/cm <sup>3</sup>   |
| Atomic weight of silicon                                         | 28.09 | g/mole              |
| $k_{\text{air}}$ (air thermal conductivity) at sea level, 20 °C  | 0.025 | W/m-K               |
| $k_{\text{air}}$ (air thermal conductivity) at sea level, 100 °C | 0.031 | W/m-K               |
| $h$ (convection heat transfer coefficient)                       | 5–10  | W/m <sup>2</sup> -K |

**Figure 5.8** Typical wafer bake profile (60-second bake followed by a 10-second transfer to a chill plate)

and bottom wafer surfaces. An example of such a full temperature profile is shown in Figure 5.8.

The impact of the time variation of temperature will depend of course on the activation energy of the reactions in question. For a nonzero activation energy, the result will be a time variation of the rate ‘constant’. For each rate constant  $k$ , an effective rate constant can be defined as

$$k_{\text{eff}} = \frac{1}{t_b} \int_0^{t_b} k(t) dt \quad (5.78)$$

where  $t_b$  is the overall bake cycle time.

It is interesting to note that the equilibrium temperature  $T^*$  is not equal to the temperature of the hot plate, as shown by Equation (5.76). The fact that the equilibrium tempera-

ture of the wafer is usually very close to the hot plate temperature indicates that heating across the gap is the dominant heat transfer mechanism. This is because the gap spacing  $\delta$  is very small, typically about 0.15–0.2 mm, so that

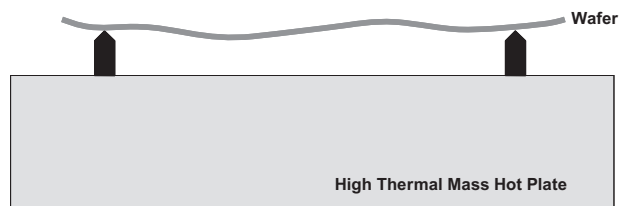
$$\text{when } \frac{k_{\text{air}}}{\delta} \gg h, \quad T^* \approx T_{\text{plate}} - \frac{\delta h}{k_{\text{air}}} (T_{\text{plate}} - T_{\text{ambient}}) \quad (5.79)$$

Even if significant variations in  $\delta$  across the wafer are present, the value of  $T^*$  will still be dominated by the temperature of the hot plate, and good temperature uniformity can be achieved. By contrast, variations in the gap spacing can have a large effect on the time constant  $\tau$ , as seen in Equation (5.77). Again, assuming  $h$  is small, the hot plate rise time will be directly proportional to  $\delta$ .

$$\text{when } \frac{k_{\text{air}}}{\delta} \gg h, \quad \tau \approx \frac{\delta \rho C_p L}{k_{\text{air}}} \quad (5.80)$$

Variations in the gap spacing can be due to misalignment of the ceramic pins that control the gap spacing in the proximity plate, or due to nonflatness of the silicon wafer – the warp of a 200- and 300-mm wafers can be up to 0.075 and 0.1 mm, respectively (see Figure 5.9).

The lumped capacitance model solution of Equations (5.75)–(5.77) has several limitations. When solving the differential Equation (5.74), all of the terms in the equation except the wafer temperature are assumed to be constant. Obviously, the accuracy of the solution will depend on the accuracy of these assumptions. As Table 5.3 indicates, the heat capacity of silicon will vary by about 10 % during a typical bake (a 100 °C temperature range or so), and the thermal conductivity of air will vary by 25 %. With a coefficient of linear expansion of about  $2.5 \times 10^{-6}/\text{K}$ , the density of silicon will change by less than 0.1 % over a 100 °C temperature range. Wafer thickness will typically vary between 1 and 2 % across the wafer. The convective heat transfer coefficient can change significantly with air flow, which will change as the air above the wafer heats up and depends on the design of the hot plate chamber. Likewise, the ambient temperature is ill-defined when the hot plate is not in an open environment and can change from wafer to wafer if the hot plate chamber lid is not temperature controlled. But possibly the most significant problem is the assumption that the hot plate temperature is constant. Just as the wafer temperature is affected by the proximity of the hot massive plate beneath it, the hot plate is affected by the cold wafer. When a new wafer is placed on a hot plate, the surface temperature of the plate can drop by 1–2 °C, requiring 5–10 seconds to heat back up to its set point temperature.



**Figure 5.9** Proximity bake of a wafer on a hot plate showing (in a highly exaggerated way) how wafer warpage leads to a variation in proximity gap (drawing not to scale)

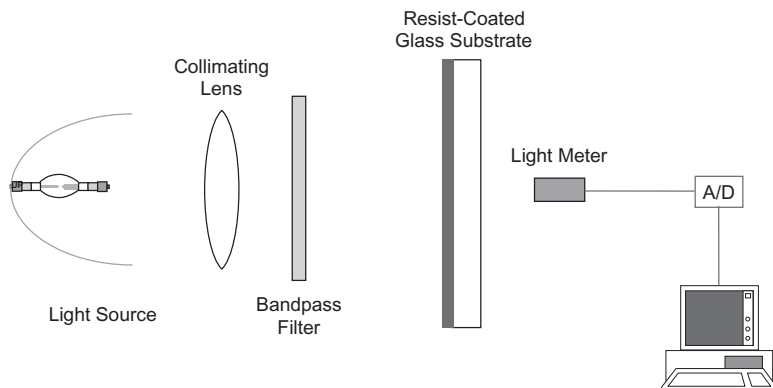
The thermal model here also assumes a one-dimensional geometry, so that variations in temperature in the plane of the wafer are ignored. Besides edge effects that can cause a temperature difference at the edge of the wafer versus the middle, wafer warpage as depicted in Figure 5.9 will lead to thermal gradients and thus heat transfer in the plane of the wafer, somewhat mitigating the temperature variations that would be predicted from a 1D model.<sup>22</sup>

## 5.5 Measuring the ABC Parameters

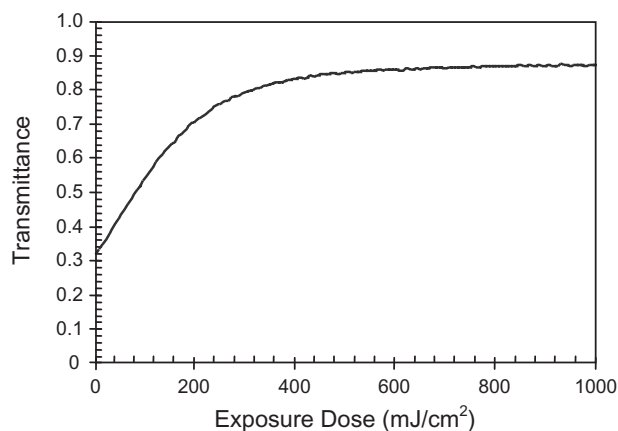
Dill proposed a single, simple experiment for measuring the *ABC* parameters.<sup>1</sup> The photoresist to be measured is coated in a nonreflecting transparent substrate (e.g. glass, quartz or similar material). The resist is then exposed by a normally incident parallel beam of light at the wavelength of measurement. At the same time, the intensity of the light transmitted through the substrate is measured continuously. The output of the experiment, transmitted intensity as a function of exposure time, is then analyzed to determine the resist *ABC* parameters. A typical experimental setup is shown in Figure 5.10. By measuring the incident exposing light intensity, the output of the experiment becomes overall transmittance as a function of incident exposure dose,  $T(E)$ . Figure 5.11 shows a typical result. Assuming careful measurement of this function, and a knowledge of the thickness of the photoresist, all that remains is the analysis of the data to extract the *ABC* parameters.

Note that the effectiveness of this measurement technique rests with the nonzero value of the resist bleachable absorption parameter *A*. If the photoresist does not change its optical properties with exposure (i.e. if  $A = 0$ ), then measuring transmittance will provide no insight into the exposure reaction, making *C* unobtainable by this method. Other methods for determining *C* for the case when  $A = 0$  are discussed briefly in Chapter 6.

Analysis of the bleaching experimental data is greatly simplified if the experimental conditions are adjusted so that the simple exposure and absorption Equations (5.29) and (5.32) apply exactly. This means that light passing through the resist must not reflect at



**Figure 5.10** Experimental configuration for the measurement of the *ABC* parameters



**Figure 5.11** Typical transmittance curve of a positive g- or i-line bleaching photoresist measured using an apparatus similar to that pictured in Figure 5.10

the resist–substrate interface. Further, light passing through the substrate must not reflect at the substrate–air interface. The first requirement is met by producing a transparent substrate with the same index of refraction as the photoresist. The second requirement is met by coating the backside of the substrate with an interference-type antireflection coating.

Given such ideal measurement conditions, Dill showed that the  $ABC$  parameters can be obtained from the transmittance curve by measuring the initial transmittance  $T(0)$ , the final (completely exposed) transmittance  $T(\infty)$  and the initial slope of the curve. The relationships are:

$$\begin{aligned}
 A &= \frac{1}{D} \ln \left[ \frac{T(\infty)}{T(0)} \right] \\
 B &= -\frac{1}{D} \ln \left[ \frac{T(\infty)}{T_{12}} \right] \\
 C &= \frac{A + B}{AT(0)[1 - T(0)]T_{12}} \left. \frac{dT}{dE} \right|_{E=0}
 \end{aligned} \tag{5.81}$$

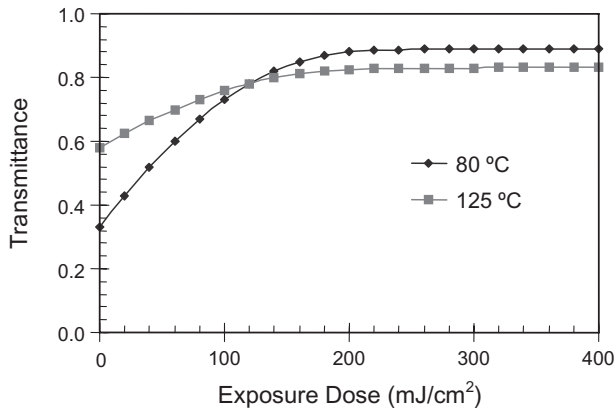
where  $D$  is the resist thickness,  $E$  is the exposure dose, and  $T_{12}$  is the transmittance of the air–resist interface and is given approximately, for a resist index of refraction  $n_{\text{resist}}$ , by

$$T_{12} = 1 - \left( \frac{n_{\text{resist}} - 1}{n_{\text{resist}} + 1} \right)^2 \tag{5.82}$$

Although graphical analysis of the data is quite simple, it suffers from the common problem of errors when measuring the slope of experimental data. As a result, the value of  $C$  (and to a lesser extent,  $A$ ) obtained often contains significant error. Dill also proposed a second method for extracting the  $ABC$  parameters from the data. Again assuming that the ideal experimental conditions had been met, the  $ABC$  parameters could be obtained by directly solving the two coupled differential Equations (5.27) and (5.32), and finding

**Table 5.4** Typical values for resist ABC parameters

| Resist Type     | $A$ ( $\mu\text{m}^{-1}$ ) | $B$ ( $\mu\text{m}^{-1}$ ) | $C$ ( $\text{cm}^2/\text{mJ}$ ) | Refractive Index        |
|-----------------|----------------------------|----------------------------|---------------------------------|-------------------------|
| g-line (436 nm) | 0.6                        | 0.05                       | 0.015                           | 1.65                    |
| i-line (365 nm) | 0.9                        | 0.05                       | 0.018                           | 1.69                    |
| 248 nm          | 0.0                        | 0.50                       | 0.05                            | 1.76                    |
| 193 nm          | 0.0                        | 1.20                       | 0.05                            | 1.71                    |
| Dyed            | Unchanged                  | Increased by 0.3–0.5       | Unchanged                       | Approximately unchanged |

**Figure 5.12** Two transmittance curves for Kodak 820 resist at 365 nm. The curves are for a convection oven post-apply bake of 30 minutes at the temperatures shown

the values of  $A$ ,  $B$  and  $C$  for which the solution best fits the experimental data. Obviously, fitting the entire experimental curve is much less sensitive to noise in the data than taking the slope at one point.

Typical values for  $ABC$  parameters for several types of resists are given in Table 5.4.

As discussed above, one impact of post-apply baking for conventional resists is the decomposition of sensitizer. Since it is the sensitizer that bleaches, a loss of sensitizer results in a decrease in the value of  $A$ . Figure 5.12 shows two example transmittance curves measured for an i-line resist baked at two different temperatures.

## Problems

- Assuming a plane wave traveling through a uniform material, derive Equation (5.5).
- For a resist film of a given thickness, there is one value of the absorption coefficient that maximizes the amount of light absorbed by the resist at the bottom. (This can be seen by looking at the extremes. When  $\alpha$  is zero, all of the incident light makes

it to the bottom, but none is absorbed. When  $\alpha$  is infinite, no light makes it to the bottom, so none is absorbed there. Thus, in between these two extremes there must be a maximum.) For a resist of thickness  $D$ , and assuming a homogenous material and a nonreflecting substrate, what value of  $\alpha$  maximizes the amount of light absorbed by the resist at the bottom?

- 5.3. An i-line resist has the following properties:

$$A = 0.85 \mu\text{m}^{-1}$$

$$B = 0.05 \mu\text{m}^{-1}$$

$$C = 0.018 \text{ cm}^2/\text{mJ}$$

$$\text{Refractive index} = 1.72$$

The resist is coated to a thickness of  $1.1 \mu\text{m}$  on a glass substrate optically matched to the photoresist.

- (a) At the beginning of exposure, what percentage of the incident light makes it to the bottom of the resist?
  - (b) The photoresist is exposed so that 55 % of the sensitizer is converted at the top of the resist. Give your best estimate of what percentage of the incident light makes it to the bottom of the resist at the end of the exposure. Explain why this estimate is not exact.
- 5.4. For most conventional (g- and i-line) resists,  $a_M \gg a_P$ . For this case, show that a measurement of bleachable absorption coefficient as a function of wavelength,  $A(\lambda)$ , can be used to determine how the exposure rate constant  $C$  varies with wavelength.
- 5.5. Derive Equations (5.35).
- 5.6. Reciprocity during exposure is seen for the case of a nonbleaching resist by transforming Equation (5.29) to Equation (5.31) – all instances of incident intensity and exposure time can be replaced by incident dose.
- (a) Show that the first Equation (5.35) obeys reciprocity.
  - (b) Show that the general first-order exposure Equation (5.27) results in reciprocity regardless of the effects of bleaching.
- 5.7. Suppose that a resist was found to produce the same thickness after PAB for a  $95^\circ\text{C}$ , 300-second bake, a  $100^\circ\text{C}$ , 60-second bake, or a  $105^\circ\text{C}$ , 17-second bake. If the DNQ of that resist decomposes with an activation energy of 31 Kcal/mole and  $\ln(Ar) = 37$  (1/minute), which bake is preferable from a thermal decomposition perspective?
- 5.8. The modified Fujita–Doolittle Equation (5.61) predicts a temperature dependence for solvent diffusivity which is not the Arrhenius equation. Using the parameters from Table 5.1 and assuming 10% solvent volume fraction, create an Arrhenius plot ( $\log [\text{diffusivity}]$  vs.  $1/\text{absolute temperature}$ ) for solvent diffusivity from  $115$  to  $130^\circ\text{C}$ . How close is this to true Arrhenius behavior? Find a best estimate for  $A_r$  and  $E_a$  based on this plot.
- 5.9. Solve the 1D diffusion equation for a ‘fixed source’ boundary condition (diffusion of a constant concentration of material at the surface of a very thick resist). Assume the diffusivity is constant and

$$C(z,0) = 0$$

$$C(0,t) = C_0$$

$$C(\infty,t) = 0$$

- 5.10. Solve the 1D diffusion equation for the 'initial source' boundary condition (diffusion of a given amount material at the surface of a very thick resist). Assume the diffusivity is constant and

$$C(z,0) = 0$$

$$dC(0,t)/dt = 0$$

$$C(\infty,t) = 0$$

$$\int_0^{\infty} C(z,t) dz = Q_0 = \text{constant}$$

- 5.11. For a typical 300-mm silicon wafer baked at a hot plate temperature of 100°C (ambient temperature of 20°C), and assuming a proximity gap of 0.2 mm, calculate:
- The equilibrium wafer temperature ( $T^*$ ) and the rise time of the bake ( $\tau$ ).
  - The equilibrium wafer temperature and the rise time of the bake if the proximity gap were increased to 0.3 mm.
  - At what proximity gap distance would the coefficient of conduction (heat transfer from the hot plate across the gap) equal that of convection (thermal transfer from the top of the wafer)?
- 5.12. For a typical 300-mm silicon wafer baked at a hot plate temperature of 100°C and a proximity gap of 0.2 mm, the equilibrium wafer temperature was found to vary by  $\pm 0.2^\circ\text{C}$  across the wafer. How much wafer warpage would be needed to explain all of this temperature variation?
- 5.13. Derive Equation (5.79).
- 5.14. From the transmittance curve in Figure 5.11, estimate the values of  $A$ ,  $B$  and  $C$ . The resist thickness used was  $0.75\ \mu\text{m}$  and the measurement was performed in the standard way. Assume a typical i-line resist.

## References

- 1 Dill, F.H., Hornberger, W.P., Hauge, P.S. and Shaw, J.M., 1975, Characterization of positive photoresist, *IEEE Transactions on Electron Devices*, **ED-22**, 445–452.
- 2 Mack, C.A., 1988, Absorption and exposure in positive photoresist, *Applied Optics*, **27**, 4913–4919.
- 3 Dammel, R., 1993, *Diazonaphthoquinone-based Resists*, SPIE Tutorial Texts, Vol. TT 11 (Bellingham, WA).
- 4 Albers, J. and Novotny, D.B., 1980, Intensity dependence of photochemical reaction rates for photoresists, *Journal of the Electrochemical Society*, **127**, 1400–1403.
- 5 Herrick, C.E., Jr., 1966, Solution of the partial differential equations describing photo-decomposition in a light-absorbing matrix having light-absorbing photoproducts, *IBM Journal of Research and Development*, **10**, 2–5.

- 6 Dill, F.H. and Shaw, J.M., 1977, Thermal effects on the photoresist AZ1350J, *IBM Journal of Research and Development*, **21**, 210–218.
- 7 Shaw, J.M., Frisch, M.A. and Dill, F.H., 1977, Thermal analysis of positive photoresist films by mass spectrometry, *IBM Journal of Research and Development*, **21**, 219–226.
- 8 Mack, C.A. and Carback, R.T., 1985, Modeling the effects of prebake on positive resist processing, *Proceedings of the Kodak Microelectronics Seminar*, 155–158.
- 9 Doolittle, A.K., 1951, Studies in Newtonian flow. II. The dependence of the viscosity of liquids on free-space, *Journal of Applied Physics*, **22**, 1471–1475.
- 10 Bueche, F., 1953, Segmental mobility of polymers near their glass temperature, *Journal of Chemical Physics*, **21**, 1850–1855.
- 11 Fox, T.G. and Flory, P.J. 1950, Second-order transition temperatures and related properties in polystyrene. I. Influence of molecular weight, *Journal of Applied Physics*, **21**, 581–591.
- 12 Fox, T.G. and Flory, P.J., 1951, Further studies on the melt viscosity of polyisobutylene, *Journal of Physical and Colloid Chemistry*, **55**, 221–234.
- 13 Fujita, H., Kishimoto, A. and Matsumoto, K., 1960, Concentration and temperature dependence of diffusion coefficients for systems polymethyl acrylate and n-alkyl acetates, *Transactions of the Faraday Society*, **56**, 424–437.
- 14 Mack, C.A., 1998, Modeling solvent effects in optical lithography, PhD Thesis, University of Texas at Austin.
- 15 Williams, M.L., Landel, R.F. and Ferry, J.D., 1955, The temperature dependence of relaxation mechanisms in amorphous polymers and other glass-forming liquids, *Journal of the American Chemical Society*, **77**, 3701–3706.
- 16 Cohen, M.H. and Turnbull, D., 1959, Molecular transport in liquids and gases, *Journal of Chemical Physics*, **31**, 1164–1169.
- 17 Ferry, J.D., 1980, *Viscoelastic Properties of Polymers*, 3rd edition, John Wiley & Sons, Ltd (New York, NY).
- 18 Mack, C.A., Mueller, K.E., Gardiner, A.B., Qiu, A., Dammel, R.R., Koros, W.G. and Willson, C.G., 1997, Diffusivity measurements in polymers, part 1: lithographic modeling results, *Proceedings of SPIE: Advances in Resist Technology and Processing XIV*, **3049**, 355–362.
- 19 Walker, E.J., 1975, Reduction of photoresist standing-wave effects by post-exposure bake, *IEEE Transactions on Electron Devices*, **ED-22**, 464–466.
- 20 Mack, C.A., DeWitt, D.P., Tsai, B.K. and Yetter, G., 1994, Modeling of solvent evaporation effects for hot plate baking of photoresist, *Proceedings of SPIE: Advances in Resist Technology and Processing XI*, **2195**, 584–595.
- 21 Smith, M.D., Mack, C.A. and Petersen, J.S., 2001, Modeling the impact of thermal history during post exposure bake on the lithographic performance of chemically amplified resists, *Proceedings of SPIE: Advances in Resist Technology and Processing XVIII*, **4345**, 1013–1021.
- 22 Tay, A., Ho, W.K. and Hu, N., 2007, An *in situ* approach to real-time spatial control of steady-state wafer temperature during thermal processing in microlithography, *IEEE Transactions on Semiconductor Manufacturing*, **20**, 5–12.